The First Counterexample Has 46 Vertices

A proposed exact solution to Ghalavand's Zagreb-index threshold question

Meheruem — Research Experiment 004

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Abstract

We determine the smallest order of a connected simple graph with minimum degree two and maximum degree five that violates the normalized Zagreb-index inequality. The first counterexample has 46 vertices. Consequently, the answer to Question I of Ghalavand [1] is 45. We give an explicit 46-vertex construction, a self-contained proof excluding every smaller graph, and a construction of counterexamples at every integer order at least 46. We also determine the unique edge-degree profile of every 46-vertex counterexample. Exact-arithmetic certificate verification, an exhaustive relaxation over edge-type counts, symbolic identities, and a separate graph-library implementation agree with the proof. The proposed contribution is the exact threshold, extremal profile, and all-orders construction—not the already-known falsity of the unrestricted Zagreb inequality.

Status. This is an unrefereed, AI-assisted research report prepared for Adnan Husain's Meheruem Substack. The argument is presented as a complete proof, but it has not been independently reviewed by a mathematician or formalized in a proof assistant. Targeted literature searches did not locate an earlier resolution of this exact threshold question; that does not establish bibliographic priority.

1. The question, in ordinary language

A graph is a collection of vertices joined by edges. The degree of a vertex is the number of edges touching it. A simple graph has neither self-loops nor repeated edges; a connected graph is one in which every pair of vertices can be joined by a path.

This investigation concerns graphs where every vertex has between two and five neighbors, with at least one vertex having two and at least one having five. We compare two averages: the average squared degree of a vertex, and the average product of the degrees at the two ends of an edge.

The original suggestion that the first average never exceeds the second was already disproved in 2007 [2]. That is not a new result of this investigation. The narrower question is: **how small can a connected counterexample be under these particular degree restrictions?**

Ghalavand's published paper [1, p. 451, Question I] asks for the largest integer up to which every such graph satisfies the inequality. Its exhibited counterexample family starts at 218 vertices. The number 218 is a feature of that construction, not a proved universal lower bound.

The answer established below is **45**. The first failure occurs at **46**. Moreover, there is a failure at every larger order.

2. Definitions and theorem

For a finite simple graph G, write $n = |V(G)|$, $m = |E(G)|$, and let $d(v)$ denote a vertex degree. Its first and second Zagreb indices are

$$M_1(G) = \sum_{v \in V(G)} d(v)^2, \qquad M_2(G) = \sum_{uv \in E(G)} d(u)d(v).$$

The normalized Zagreb inequality is

$$\frac{M_1(G)}{n} \leq \frac{M_2(G)}{m}.$$

Throughout, define the integer gap

$$D(G) = mM_1(G) - nM_2(G).$$

Thus a counterexample has $D(G) > 0$. Let e_{ij} be the number of edges whose endpoint degrees are i and j, with $i \leq j$.

Theorem. Among finite connected simple graphs with minimum degree two and maximum degree five:

1. Every graph with at most 45 vertices satisfies the normalized Zagreb inequality.
2. For every integer $n \geq 46$, a graph of order n violating that inequality exists.
3. Every counterexample of order 46 has precisely the following nonzero edge-type counts:

$$e_{25} = 35, \qquad e_{33} = 31, \qquad e_{23} = 1.$$

Its vertex-degree counts and invariants are therefore

$$(n_2, n_3, n_4, n_5) = (18, 21, 0, 7),$$

$$(n, m, M_1, M_2, D) = (46, 67, 436, 635, 2).$$

In particular, the answer to Question I in [1] is 45. The profile in part 3 does not assert that the extremal graph is unique up to isomorphism.

3. An explicit 46-vertex counterexample

The construction joins two pieces by a single bridge.

The first piece. Label the vertices of a complete graph on seven vertices by $0, 1, \ldots, 6$. Remove the four edges

$$\{0,1\}, \quad \{0,2\}, \quad \{3,4\}, \quad \{5,6\}.$$

Seventeen edges remain. Vertex 0 has degree four and the other six vertices have degree five. The remaining graph is connected: every vertex has degree at least four, which rules out two components on seven vertices.

Subdivide each of the 17 remaining edges once, using a different new vertex for each edge. Subdivision replaces an edge by a two-edge path. Add one further vertex u, adjacent to vertex 0. This piece has 25 vertices. Its seven original vertices all have degree five, its 17 subdivision vertices have degree two, and u currently has degree one.

The second piece. Take the ten-prism $C_{10}\square K_2$: two ten-cycles joined by a matching between corresponding vertices. This is a connected cubic graph on 20 vertices with 30 edges. Subdivide one edge by a new vertex w. The resulting piece has 21 vertices and 31 edges. Its old vertices have degree three and w currently has degree two.

Join the pieces. Add the edge uw. Now u has degree two and w has degree three. The graph is simple and connected, and its minimum and maximum degrees are two and five.

The first piece supplies 35 edges of type $(2,5)$, the second supplies 31 of type $(3,3)$, and the bridge supplies one of type $(2,3)$. Consequently,

$$M_1 = 18 \cdot 2^2 + 21 \cdot 3^2 + 7 \cdot 5^2 = 436,$$

$$M_2 = 35 \cdot (2 \cdot 5) + 31 \cdot (3 \cdot 3) + 1 \cdot (2 \cdot 3) = 635.$$

The violation is exact:

$$67 \cdot 436 - 46 \cdot 635 = 2 > 0,$$

or, equivalently,

$$\frac{436}{46} - \frac{635}{67} = \frac{1}{1541} > 0.$$

No floating-point tolerance is involved. A complete, labeled edge list is included in the accompanying certificates.

4. Why no smaller counterexample exists

The proof uses the established edge-type expansion underlying prior Zagreb-index work [1], [2], [3]. The expansion itself is not claimed as new. The threshold argument comes from bounding the cost of connecting the two favorable edge types and then enforcing degree integrality.

4.1. An exact quadratic identity

Let $T = \{(i,j) : 2 \le i \le j \le 5\}$. For $t = (i,j)$ define

$$q_t = ij, \qquad r_t = \frac{1}{i} + \frac{1}{j}, \qquad s_t = i + j = q_t r_t.$$

Direct counting gives

$$n = \sum_{t \in T} r_t e_t, \qquad m = \sum_{t \in T} e_t,$$

$$M_1 = \sum_{t \in T} s_t e_t, \qquad M_2 = \sum_{t \in T} q_t e_t.$$

For example, in the first identity each vertex of degree i contributes $1/i$ through each of its i incident edges, hence contributes one overall.

For two types set

$$\Psi(t,u) = q_t r_u + q_u r_t - s_t - s_u = (q_t - q_u)(r_u - r_t).$$

Expanding the products yields

$$D = -\sum_{t<u} \Psi(t,u) e_t e_u. \tag{1}$$

The sum is over distinct **unordered** type pairs, using any fixed ordering. Diagonal coefficients vanish; no factor of two is missing.

Here are the ten types ordered by their products:

Type	q_t	r_t
$(2,2)$	4	1
$(2,3)$	6	5/6
$(2,4)$	8	3/4
$(3,3)$	9	2/3
$(2,5)$	10	7/10
$(3,4)$	12	7/12
$(3,5)$	15	8/15
$(4,4)$	16	1/2
$(4,5)$	20	9/20
$(5,5)$	25	2/5

The reciprocal sums decrease strictly except for the inversion between $(3,3)$ and $(2,5)$. Thus, of the 45 coefficients for distinct pairs, exactly one is negative:

$$\Psi((2,5),(3,3)) = -\frac{1}{30}.$$

All the other 44 are strictly positive. A graph with $D > 0$ must therefore contain both of these types. Put $a = e_{25} > 0$ and $b = e_{33} > 0$. Call the other eight types *defect types* for the purposes of this proof.

4.2. A small-order graph can afford very few connecting types

For each defect type t, let $A_t = \Psi((2,5),t)$ and $B_t = \Psi((3,3),t)$. Their exact values are:

Defect type	A_t	B_t
$(2,2)$	$9/5$	$5/3$
$(2,3)$	$8/15$	$1/2$
$(2,4)$	$1/10$	$1/12$
$(3,4)$	$7/30$	$1/4$
$(3,5)$	$5/6$	$4/5$
$(4,4)$	$6/5$	$7/6$
$(4,5)$	$5/2$	$143/60$
$(5,5)$	$9/2$	$64/15$

Define $A = \sum_t A_t e_t$, $B = \sum_t B_t e_t$, and let $C \geq 0$ denote the sum of all defect–defect contributions in (1). Then

$$D = \frac{ab}{30} - Aa - Bb - C. \tag{2}$$

Suppose $n \leq 46$, and write $S = a + b$. Since

$$n \geq \frac{7}{10}a + \frac{2}{3}b \geq \frac{2}{3}S,$$

we have $0 < S \leq 69$. If both $A \geq 7/12$ and $B \geq 7/12$, then

$$D \leq \frac{S^2}{120} - \frac{7}{12}S = \frac{S(S-70)}{120} < 0. \tag{3}$$

So a counterexample of this order cannot have both coefficient totals that large.

An edge of type $(2,2)$, $(3,5)$, $(4,4)$, $(4,5)$, or $(5,5)$ would alone exceed these thresholds. All five types are excluded. Only $(2,3)$, $(2,4)$, and $(3,4)$ remain as possible defect types.

If there is an edge of type $(2,3)$ and any second defect edge, the table gives

$$A \geq \frac{8}{15} + \frac{1}{10} = \frac{19}{30} > \frac{7}{12}, \qquad B \geq \frac{1}{2} + \frac{1}{12} = \frac{7}{12},$$

contradicting (3). Thus this possibility leaves exactly one $(2,3)$ edge and no other defect edges.

Otherwise, let $p = e_{24}$ and $q = e_{34}$. Connectedness forces $p, q \geq 1$: without both types, the degree-three vertices cannot connect to the vertices supporting the $(2,5)$ edges. Moreover,

$$p + q = 4n_4 \geq 4.$$

If $q \geq 2$, these conditions imply $A, B \geq 2/3$. Indeed, either $p, q \geq 2$, or $p = 1$ and $q \geq 3$, and the table gives the bound in both cases. If $q = 1$ and $p \geq 4$, then $A \geq 19/30$ and $B \geq 7/12$. Both alternatives contradict (3).

There are therefore only two possible defect patterns:

$$\text{I: } e_{23} = 1; \qquad \text{II: } e_{24} = 3,\ e_{34} = 1.$$

All defect counts not displayed are zero.

4.3. Pattern II is impossible at order at most 46

For pattern II, direct substitution gives

$$n = \frac{7}{10}a + \frac{2}{3}b + \frac{17}{6},$$

$$D = \frac{ab}{30} - \frac{8}{15}a - \frac{1}{2}b - 2.$$

The constant two is the cross contribution $3\Psi((2,4),(3,4)) = 3(2/3)$. Positivity of D forces $a > 15$, $b > 16$, and

$$(a - 15)(b - 16) > 300.$$

Set $X = 21(a - 15) > 0$ and $Y = 20(b - 16) > 0$. The order bound implies

$$21a + 20b + 85 \leq 1380, \qquad X + Y \leq 660.$$

But then

$$XY \leq \left(\frac{X + Y}{2}\right)^2 \leq 108900,$$

whereas positivity requires $XY > 420 \cdot 300 = 126000$. This contradiction excludes pattern II.

4.4. Pattern I forces the threshold and the extremal profile

For pattern I,

$$n = \frac{7}{10}a + \frac{2}{3}b + \frac{5}{6}, \qquad D = \frac{ab - 16a - 15b}{30}. \tag{4}$$

Counting incident edges at vertices of each degree gives

$$a = 5n_5, \qquad a + 1 = 2n_2, \qquad 2b + 1 = 3n_3.$$

Hence

$$a \equiv 5 \pmod{10}, \qquad b \equiv 1 \pmod{3}.$$

Since $D > 0$, we have $a > 15$ and $b > 16$, so the congruence gives $b \geq 19$. The order bound in (4) now yields $a < 47$. Therefore only $a = 25, 35, 45$ need be considered.

a	Positivity requires	Smallest allowed b	Resulting minimum order
25	$b > 40$	43	47
35	$b > 28$	31	46
45	$b > 24$	25	49

The only possibility at order at most 46 is $a = 35$, $b = 31$, and $n = 46$. Formula (4) gives $D = 2$. This proves both the exclusion of all smaller counterexamples and the unique edge-degree profile at order 46.

There is also an immediate structural consequence: every extremal counterexample has one bridge separating its 25 degree-two-or-five vertices from its 21 degree-three vertices. This follows because the sole $(2, 3)$ edge is the only connection between those two nonempty vertex classes. It does not enumerate the possible internal graphs on either side.

5. Counterexamples at every order from 46 onward

The finite example extends to an explicit two-parameter family. Let $k \geq 3$ be odd and let $\ell \geq 3$.

Begin with a cycle on k hubs, labeled $0, 1, \ldots, k-1$, and double each cycle edge. Add the matching

$$\{1, 2\}, \{3, 4\}, \ldots, \{k-2, k-1\}.$$

This intermediate object is a connected loopless **multigraph**. Hub 0 has degree four and every other hub has degree five. Subdivide every individual edge using a distinct new vertex, then add a vertex u adjacent to hub 0. Because different parallel edges receive different subdivision vertices, the subdivided graph is **simple**.

Separately, take the cubic prism $C_\ell \square K_2$ and subdivide one edge with a new vertex w. Add the bridge uw. The resulting graph $G_{k,\ell}$ is simple and connected, with minimum degree two and maximum degree five.

Counting vertices and edges gives

$$n = \frac{7k + 4\ell + 3}{2}, \qquad m = 5k + 3\ell + 2,$$

$$n_2 = \frac{5k + 1}{2}, \qquad n_3 = 2\ell + 1, \qquad n_4 = 0, \qquad n_5 = k,$$

$$e_{25} = 5k, \qquad e_{33} = 3\ell + 1, \qquad e_{23} = 1.$$

It follows that

$$M_1 = 35k + 18\ell + 11, \qquad M_2 = 50k + 27\ell + 15,$$

and therefore

$$D(G_{k,\ell}) = \frac{k\ell - 5k - 3\ell - 1}{2}. \tag{5}$$

The graph is a counterexample precisely when $(k-3)(\ell-5) > 16$.

For an **even** integer $n \geq 46$, choose

$$k = 7, \qquad \ell = \frac{n-26}{2} \geq 10.$$

Equation (5) becomes $D = n - 44 > 0$.

For an **odd** integer $n \geq 47$, choose

$$k = 5, \qquad \ell = \frac{n-19}{2} \geq 14.$$

Equation (5) becomes $D = (n-45)/2 > 0$. The two cases cover every integer order at least 46 and finish the theorem's proof.

6. Reproducibility and attempts to falsify the result

The proof above is not conditional on software correctness. Computation was used to check the construction, catch arithmetic errors, and challenge the classification.

Concrete certificates. Seven labeled graphs are supplied: the explicit complete-graph construction at order 46 and family examples at orders 46, 47, 48, 49, 100, and 101. The verifier reconstructs adjacency from the edge lists. It checks endpoint validity, absence of loops and duplicate edges, connectedness, actual degrees, both Zagreb indices, the exact integer gap, and agreement with the supplied metadata. It does not trust reported degree sequences or invariant values.

An exhaustive edge-type relaxation. This is not an enumeration of all graphs. For a counterexample of order at most 46, the table shows that each defect edge contributes at least $1/10$ to A and at least $1/12$ to B. Seven defect edges would make $A \geq 7/10$ and $B \geq 7/12$, contradicting (3). Thus at most six defect edges need be considered.

The program enumerates all 3,003 nonnegative eight-coordinate defect vectors with coordinate sum at most six, together with the rigorously bounded positive counts $a \leq 65$ and $b \leq 69$. It enforces integer degree counts, the order bound, and connectedness of the degree-class support. The last condition is necessary but not sufficient for graph realizability; retaining unrealizable vectors makes the search a safe relaxation rather than an unsupported graph-generation claim.

There are 49,380 integral candidate vectors of order at most 46, of which 36,641 have connected degree-class support. Exactly one of those has positive gap: the profile proved above. No positive-gap vector has order at most 45.

Regression and negative testing. All 18 test methods pass. They include deliberately corrupted certificates, duplicate and reversed edges, self-loops, noninteger and boolean endpoints, a missing bridge, false metadata, and a valid degree-bounded graph that does not violate the inequality. The suite also verifies 1,000 exact quadratic-identity instances, 435 parameter pairs in the general family, every order from 46 through 1,000, and the additional orders 10,000 and 10,001. These finite checks do not replace the infinite proof.

A second implementation. NetworkX recomputes the seven certificates' connectivity and invariant values. SymPy verifies the two minimality-case expansions and the family gap identity. This cross-check ran with Python 3.13.5, NetworkX 3.6.1, and SymPy 1.14.0. It is a second implementation within this investigation—not independent expert review.

The main constructor, verifier, and test suite require only Python 3.10 or later and its standard library. From the extracted verification directory:

```
python verify_results_004.py example_certificates.json --exhaustive
python test_verifier_004.py
```

To regenerate the certificates:

```
python construct_graphs_004.py
```

The optional second implementation additionally requires NetworkX and SymPy.

7. What is new, and what is not claimed

The proposed new contribution is the **exact connected threshold under the specified degree restrictions**, the necessary edge profile of every smallest counterexample, and explicit counterexamples at **every** order beyond the threshold.

The unrestricted inequality was already false [2]. Ghalavand also recounts a historical connected example with 46 vertices and 110 edges [1, p. 443]; the matching vertex count must not be advertised as a first 46-vertex counterexample in the unrestricted problem. The present extremal profile has 67 edges.

The edge-type expansion, its factored coefficient, and constructions that join incompatible degree patterns are established prior methods. In particular, Andova and coauthors give a general connected construction from complete bipartite pieces [3, Proposition 3.1]. No novelty is claimed for those underlying ideas.

The stronger claim that this exact threshold has never appeared anywhere remains a bibliographic question. The search checked the original question, its journal version and preprint, relevant older primary literature, and combinations of the author's name, title, degree restrictions, and proposed threshold. No matching resolution was located. The full text of the 2009 Sun–Chen paper was not accessible during this investigation; its bibliographic record and relevance were identified through the sources reviewed. Unindexed, inaccessible, differently worded, or very recent work may have been missed.

The enumeration question in [1, Question II] is **not solved** here. An edge-degree profile is not a count of nonisomorphic graphs. No formal proof-assistant certificate or external referee report is supplied.

Publication status: a complete proposed solution with a proof and reproducible checks, awaiting independent scrutiny.

8. Investigation record

Problem selection was part of the experiment rather than fixed in advance. TxGraffiti's collection [5] was screened, but several candidates had uncertain current status or search hits reporting subsequent resolutions; none is claimed solved here. A zero-forcing discrepancy conjecture in [4, Conjecture 5.4] was also investigated. A finite check of all 1,044 seven-vertex graphs in the NetworkX Graph Atlas found no counterexample. That check was exploratory and does not settle the conjecture.

The Zagreb threshold was selected because its precise hypotheses admitted an exact local description in terms of ten edge types. The productive reduction was to separate the single favorable pair of types from all connecting penalties. A concrete 46-vertex witness then suggested the sharp threshold; the subsequent lower-bound argument established that threshold without assuming the witness was optimal.

References

[1] Ali Ghalavand. **On a Conjecture about Comparing the First and Second Zagreb Indices of Graphs.** *MATCH Communications in Mathematical and in Computer Chemistry* **96** (2026), 441–452. DOI: 10.46793/match.96-2.04725. The target is Question I on p. 451; the paper's exhibited family starts at order 218. Published paper. Preprint, arXiv:2509.07434.

[2] Pierre Hansen and Damir Vukičević. **Comparing the Zagreb Indices.** *Croatica Chemica Acta* **80**(2) (2007), 165–168. Establishes the historical failure of the unrestricted inequality and validity for chemical graphs. Publisher record and article.

[3] Vesna Andova, Sašo Bogoev, Darko Dimitrov, Marcin Pilipczuk, and Riste Škrekovski. **On the Zagreb index inequality of graphs with prescribed vertex degrees.** *Discrete Applied Mathematics* **159**(8) (2011), 852–858. DOI: 10.1016/j.dam.2011.01.002. Prior work on the factored edge-type coefficient, good degree intervals, and connected counterexample constructions. Author-hosted preprint.

[4] Andrew McKay, Allie Ray, and Jonathan Valliere. **Maximizing the discrepancy between zero forcing parameters relative to graph order.** arXiv:2609.35522v1 (28 September 2026). Conjecture 5.4 was a screened alternative, not a claimed result of this report. Preprint.

[5] Randy Davila, Boris Brimkov, and Ryan Pepper. **In Reverie Together: Ten Years of Mathematical Discovery with a Machine Collaborator.** arXiv:2507.17780v1 (23 July 2025). A collection used during candidate selection. Preprint.